

Yasiru Laksara

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[Yasiru Laksara](#) | [YasiruLaksara](#) | [yasirulaksara.github.io](#)

Dept. of Computer Science and Engineering, University of Moratuwa, Sri Lanka



OBJECTIVE

I am a self-motivated Computer Science and Engineering undergraduate at the University of Moratuwa, dedicated to becoming a qualified professional in the field. I am particularly interested in Data Science, Machine Learning, and Web Development, and I am committed to continuous learning and effective teamwork. I aim to apply my growing expertise to make meaningful contributions to the fields of machine learning and artificial intelligence.

EXPERIENCE

CML Insight Inc.

Machine Learning Engineering Intern

Dec 2024 – May 2025

Remote / Sri Lanka

- Developed intelligent multi-agent systems for a Causal AI-Driven Education Intervention System, utilizing OpenAI Agent SDK and Google ADK to support adaptive human-AI collaboration.
- Engineered autonomous pipelines for LLM performance testing using frameworks like DeepEval and gained hands-on exposure to fine-tuning OpenAI-based models tailored to specific use cases.
- Developed a custom function-based plugin to enhance document extraction workflows for unstructured research data, optimizing system efficiency and reducing API costs.
- Conducted technical research on agentic architectures and delivered a Tech Talk on OpenAI Custom Functions to share implementation strategies with the engineering team.

EDUCATION

University of Moratuwa

B.Sc. (Hons) in Computer Science and Engineering, specializing in Data Science and Engineering

2022 – Present

Katubedda, Sri Lanka

- GPA: 3.67/4.00

H/Vijayaba National School Hungama

GCE Advanced Level Examination – Physical Science Stream

Oct 2020

Hungama, Sri Lanka

- Z-score: 2.0939

- Combined Mathematics – A / Physics – A / Chemistry – A / General English – A

H/Vijayaba National School Hungama

GCE Ordinary Level Examination

Dec 2017

Hungama, Sri Lanka

- 8 As and 1 B (B in English Literature)

PROJECTS

NeuraShield — Adaptive Web Defense System

Jul 2025 – Present

Tools: Python, FastAPI, PostgreSQL, Hugging Face Transformers, ONNX Runtime

- An adaptive, real-time web defense system utilizing Continual Learning and ONNX-optimized transformers to detect sophisticated threats such as SQLi, XSS, SSRF, and OS Command Injection. Features Explainable AI (XAI) for transparent decision making and a Human-in-the-Loop mechanism for continuous model refinement.

InnoVest — Microservices-Based Innovation Funding Platform

Jan 2025 – Jun 2025

Tools: Spring Boot, Keycloak, Kafka, Flyway, Kubernetes, Docker, ArgoCD, Prometheus, Grafana, Stripe, GCS

[GitHub Repo](#)

- Built a web-based platform connecting innovators with investors through a transparent bidding system. Streamlined the funding process via automated interest based matching, detailed product showcases, and secure payment integration for subscriptions.

ResQ — AI-Powered Disaster Response Coordination Platform

May 2025

Tools: LangChain, LangGraph, LangSmith, FastAPI, PostgreSQL, Firebase, GCP, Gemini, DeepEval, Tavily API

[GitHub Repo](#)

- Developed an AI-powered disaster response platform that processes multimodal inputs (text, voice, and images) to capture urgent requests and route them intelligently. Built a priority based matching engine that assigns responders based on location, skills, and urgency, with role specific dashboards for victims, volunteers, first responders, and government agencies to ensure coordinated, safe, and efficient disaster relief operations.

HealthBot+ — AI-Powered Web Application for Early Skin Disease Detection

Jul 2024 – Sep 2024

Tools: React.js, Flask, Node.js, MongoDB, Firebase, Vercel, Hugging Face, Grad-CAM, OpenAI API

[GitHub Repo](#)

- Developed an AI-powered web application for early skin disease detection using transfer learning and Explainable AI, integrating a medical chatbot and automated reporting for accurate and transparent patient care.
- **T20-Cricket Score Predictor** Jun 2024 – Dec 2024
Tools: Python, Flask, HTML, CSS [GitHub Repo](#)
◦ Developed a web-based T20 Cricket Score Predictor using machine learning to estimate final scores based on current match data and game context.
- **Automatic Number Plate Recognition** June 2024
Tools: Python, OpenCV, Ultralytics, YOLOv8 [GitHub Repo](#)
◦ Implemented an ANPR system to detect and read license plates in video files, visualizing results with bounding boxes.
- **HR Management System** Aug 2023 – Nov 2023
Tools: Node.js, MySQL, Express.js, React [GitHub Repo](#)
◦ Developed as part of a Semester 3 group project, this full-stack HRMS featured robust database structures and key backend functionalities to manage employee data and HR operations. I led database creation and developed critical backend components, ensuring the system's effectiveness.
- **NATOURS — RESTful API for Tour Management** Jul 2024 – Jan 2025
Tools: Node.js, Express.js, MongoDB, JWT, Mongoose [GitHub Repo](#)
◦ Built a RESTful API backend for managing tour operations, including tour creation, user authentication, and booking functionalities.
- **RPAL Interpreter — Lexical Analyzer and Parser** 2023
Tools: C++ [GitHub Repo](#)
◦ As part of the Programming Languages module, developed a lexical analyzer and parser for RPAL, including AST generation and a Control Stack Environment machine for interpreting RPAL code.

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [S.1] Yasiru Laksara and Uthayasanker Thayasivam (2025). **Enhancing Multi-Label Thoracic Disease Diagnosis with Deep Ensemble-Based Uncertainty Quantification**. Submitted to *ICITR 2025* (under peer review).
Preprint available at: [arXiv:2511.18839](https://arxiv.org/abs/2511.18839)

SKILLS

- **Programming Languages:** Python, Java, C++
- **Web Technologies:** HTML, CSS, JavaScript, React.js, Node.js, Spring Boot, FastAPI, Flask, Streamlit
- **Database Systems:** PostgreSQL, MySQL, MongoDB
- **Data Science & Machine Learning:** NumPy, Pandas, scikit-learn, TensorFlow, Keras, OpenCV, Hugging Face Transformers, ONNX, Matplotlib, Seaborn
- **Cloud Technologies:** Google Cloud Platform, Firebase, Netlify
- **DevOps & Version Control:** Git, GitHub
- **Agentic / LLM Frameworks:** Google ADK, OpenAI SDK, LangChain, LangGraph, LangSmith, LangFuse

HONORS AND AWARDS

- **1st Place (Champions): CYPHER 2.0 Hackathon 2024** March 2024
IEEE WIE Student Branch Affinity Group of KDU 
◦ Showcased problem-solving and time management skills in a 5-hour coding challenge.
- **1st Runner-Up: IntelliHack 5.0** June 2025
IEEE Computer Society Student Branch Chapter of UCSC 
◦ Developed an agent-based system to improve disaster response coordination, competing against over 100 teams and 350+ participants.
- **2nd Runner-Up: ML Olympiad – Sustainable Urban Living** March 2024
Kaggle 
◦ Demonstrated machine learning problem-solving skills by individually developing a model to predict property habitability scores, distinguishing myself among 120+ participants.
- **2nd Runner-Up: CodeFest-2024 Datathon Competition** Jan 2025
SLIIT
◦ Applied machine learning, data analysis, and problem-solving skills throughout a 6-hour datathon challenge.
- **9th Place: HaXtreme 3.0** Oct 2024
IEEE Computer Society, University of Ruhuna 
◦ Showcased problem-solving and time management skills in a 6-hour coding challenge, excelling among 100+ competing teams.

VOLUNTEER EXPERIENCE

• Editorial Committee Member <i>Mora UXplore 2.0 – IEEE Student Branch, University of Moratuwa</i>	<i>Feb 2024 – Dec 2024</i>
• Finance Committee Member <i>M-Tutor – Maths Society, University of Moratuwa</i>	<i>Feb 2024 – Dec 2024</i>
• Editorial Committee Member <i>NEXTGEN 1.0 – IEEE Robotics & Automation Society, University of Moratuwa</i>	<i>Jan 2024 – Apr 2024</i>
• Publicity Committee Member <i>SLIOT Challenge – Department of CSE, University of Moratuwa</i>	<i>Oct 2023 – Apr 2024</i>
• Editorial Committee Member <i>MoraXtreme 8.0 – IEEE Computer Society</i>	<i>Oct 2023 – Jan 2024</i>
• Event Committee Member <i>EXMO 2023, the flagship technological exhibition of the University of Moratuwa</i>	<i>Jul 2023</i>
• Project Member <i>Shilpasara – LEO Club of the University of Moratuwa</i>	<i>Oct 2022 – Nov 2023</i>
• Organizing Committee Member <i>CSE Career Fair 2024 – Department of CSE, University of Moratuwa</i>	<i>Jan 2023</i>

PROFESSIONAL MEMBERSHIPS

• Student Member, Institution of Engineers, Sri Lanka (IESL) , Membership ID: S-30305	<i>2024 – Present</i>
• Member, IEEE Student Branch, University of Moratuwa, Sri Lanka	<i>2023 – Present</i>

CERTIFICATIONS

• Fundamentals of Deep Learning	<i>Sep 24, 2025</i>
• Certifying Body: NVIDIA	
• Machine Learning Foundations	<i>Feb 15, 2024</i>
• Certifying Body: AWS Academy	
• Supervised Machine Learning	<i>Apr 8, 2024</i>
• Certifying Body: Coursera	
• Intro to Machine Learning	<i>Jan 26, 2024</i>
• Certifying Body: Kaggle	

ADDITIONAL INFORMATION

Languages: English (Fluent), Sinhala (Native)
Interests: Sports (Athletics, Cricket), Photography

REFERENCES

- Akila Peiris**
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